

MADHAV KRISHNAMANGALAM SABU

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Machine learning engineer with mechanical background focusing on evaluation and control of physical systems that cannot be experimented on directly. Built a half million transition dataset for offline reinforcement learning and 3-stage closed loop harness that showed the field's standard evaluation protocol certifies policies that fail in real deployment; my own tuned policy lost to a trivial fixed setpoint, for which the findings are under review for NeurIPS 2026 workshop. Have 3+ years of experience evaluating and instrumenting real world thermal systems along with prior experience on atmospheric modeling using 200+ radiosonde profiles and 50,000+ hourly temperature records.

MACHINE LEARNING RESEARCH

Offline Reinforcement Learning for Price-Responsive HVAC Control

2026

Independent Research; paper under review, CCAI Workshop NeurIPS 2026

Python, PyTorch, NYISO Market Data, EnergyPlus

- Built a 523,775-transition offline RL dataset with 5 min EnergyPlus simulation data and 2023 NYISO Zone J real-time and day-ahead price data, merged it into a 35-50 feature MDP state across 1,825 episodes and 5 logged behavior policies.
- Trained and benchmarked discrete-action CQL agents, behavior cloning, FQI and a learned one-step dynamics model with $R^2 = 0.960$ and next temperature MAE = 0.147 °C.
- Designed a 3-stage evaluation harness of increasing fidelity: learned dynamics rollout, open loop simulator replays and a closed loop replay reading true zone temperatures back into the state every 5 mins.
- Closed loop evaluation showed a trivial 24 °C setpoint achieved the lowest cost and comfort penalty across all 18 cooling-active episodes, beating every learned policy.
- Diagnosed the safety router failure to control logic rather than forecasting by isolating a one-line defect in the recovery branch; the fix reduced realized comfort penalty 6x, but the result after re-verification remained unchanged.

Fleet-Scale Battery Degradation Anomaly Detection

2026

Open-source portfolio project, in progress

Python, DuckDB, PyTorch

- Constructed an ingestion and validation pipeline over the Severson fast-charging dataset creating a modeling set of 139 cells and ~114,000 cell-cycle records in Parquet (queryable through DuckDB).
- Identified physically impossible near-zero internal resistance readings clustered at early cycles and end of life, retained and flagged as sensor artifacts.
- Implementing a discrete-time hazard model with conformal prediction for calibrated uncertainty rather than a naïve time to failure regression by characterizing a ~70% right-censoring structure as the core modeling constraint.

Atmospheric and Urban Heat Transfer Modeling, Columbia University

2023

Research with Prof. Arvind Narayanaswamy

Python, Regression Modeling

- Analyzed 50,000+ hourly temperature records to model seasonal and diurnal heat transfer behavior in urban environments.
- Applied conduction, convection, and radiation heat transfer models combined with regression analysis to quantify impacts of urbanization variables on thermal response.
- Integrated 200+ radiosonde atmospheric profiles to simulate gas-specific heat transfer effects, improving predictive accuracy of temperature rise and thermal intensity.

PROFESSIONAL EXPERIENCE

Empowered Buildings

New York, USA

Senior Energy Engineer

October 2024 - Present

- Analyzed 20M+ kWh of interval electricity data and 1.2M+ therms of gas data across a building portfolio, building Python and spreadsheet models that reconcile predicted against measured performance and quantify where models diverge from instrumented reality.
- Designed and built an internal retrieval-augmented generation system (FastAPI, React, LangChain, ChromaDB) giving the team a citation-backed chat interface over utility and state incentive program rules, with deterministic savings and incentive calculations rather than model-generated arithmetic.
- Led system-level thermal and energy performance analysis across retrofit projects, evaluating HVAC and distribution behavior under real operating conditions.

En-Power Group

New York, USA

Energy Engineer I | Engineering & Auditing Intern

May 2023 – October 2024

- Performed investment-grade analysis for 20+ buildings and served as lead auditor for 9 multifamily properties (1,000+ units, 2M+ sq ft), directing on-site instrumentation, data collection and system diagnostics.
- Conducted field audits across 15+ buildings, including daylighting, sub-metering and load analyses, and validated modeled outputs against observed system behavior for 20+ regulatory filings.

EDUCATION

Columbia University

New York, USA

Master of Science, Mechanical Engineering,

Sept 2022 - Dec 2023

Relevant Coursework: Energy Economics and Optimization, Data science for Mechanical Engineers, Building Energy and Simulation, Advanced Heat Transfer, Advanced Fluid Mechanics.

National Institute of Technology

Tiruchirappalli, India

Bachelor of Technology, Mechanical Engineering,

Aug 2018 - May 2022

Relevant Coursework: Thermal Engineering, Heat and Mass Transfer, Computational Fluid Dynamics, Fluid Mechanics, Data Structures and Algorithms, Database Management Systems.

SKILLS & INTERESTS

Machine learning: Offline RL (Conservative Q-Learning, fitted Q-iteration, behavior cloning), discrete-time hazard models, conformal prediction, anomaly detection, time-series forecasting, closed-loop policy evaluation

Languages & libraries: Python (PyTorch, NumPy, Pandas, SciPy, scikit-learn), MATLAB, C (read), SQL

Data & systems: DuckDB, Parquet pipelines, large heterogeneous scientific datasets, FastAPI, React, LangChain, ChromaDB, Git

Scientific modeling: EnergyPlus, COMSOL, ANSYS, SolidWorks, heat transfer, field instrumentation, IR thermography